Kalaignar Karunanidhi Institute of Technology
Department of Computer Science and Engineering

PROJECT REPORT

BOLTZMANN MACHINE

Subject: Deep Neural Networks (DNN)

Submitted by

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Abstract

A Boltzmann Machine is a stochastic neural network that learns the probability distribution of its input data. It consists of binary units connected by symmetric weights, and every state of the network is given an energy value. States with low energy are more probable than states with high energy. During learning, the weights and biases are adjusted so that the patterns in the training data receive low energy. Training the general model is very slow, so the Restricted Boltzmann Machine (RBM), which removes the connections within each layer, is used more often and is commonly trained with an approximate method called Contrastive Divergence. Boltzmann Machines have been applied to feature learning, recommendation systems, pattern recognition, generative modelling and dimensionality reduction. This report explains their architecture, energy function, learning process, types, applications, advantages and limitations.

1. Introduction

A Boltzmann Machine is a neural network whose units take binary values (0 or 1) and are joined to each other by symmetric weights. It was introduced by Geoffrey Hinton and Terrence Sejnowski in the early 1980s and described in detail by Ackley, Hinton and Sejnowski (1985). The name comes from the Boltzmann distribution of statistical physics, which relates the probability of a state to its energy. It can be seen as a stochastic version of the Hopfield network, which also uses symmetric weights but updates its units in a fixed, deterministic way. Boltzmann Machines are important in neural network research because they showed that a network can learn useful internal representations from unlabelled data. Later, Restricted Boltzmann Machines were stacked to build deeper models, which was an important step in the development of deep learning.

1.1 Basic Concept

Stochastic neural networks. In an ordinary feed-forward network, the same input always gives the same output. In a stochastic network, the state of each neuron is decided with some probability, so the network can reach different states on different runs. This randomness lets the network explore many possible states instead of getting stuck in one.

Energy-based models. An energy-based model gives every possible state of the network a single number called its energy. A state that agrees with the learned patterns has low energy, and an unlikely state has high energy. The probability of a state is calculated from its energy, so low-energy states are the most probable. Learning therefore means shaping the energy so that the training data have low energy.

Learning a distribution and predicting labels. A feed-forward classifier learns a mapping from an input to a labelled output. A Boltzmann Machine instead models how the input data themselves are distributed, without needing labels. The features it learns can later be given to a classifier, but a Boltzmann Machine by itself is not a direct label predictor.

1.2 Objective

The objective of studying Boltzmann Machines is to understand how a neural network can learn a probability distribution from unlabelled data, how the energy function defines this distribution, and how the weights are adjusted during training. The report also compares the general, restricted and deep forms of the model and discusses their applications and limitations.

2. Architecture and Working Principle

2.1 Architecture

A Boltzmann Machine is made of binary units, where the state s_i of unit i is 0 or 1. The units are divided into **visible units**, which receive the data from outside, and **hidden units**, which are not observed directly and learn the dependencies between the visible units. In a general Boltzmann Machine, every unit is connected to every other unit by an undirected connection (Figure 1). Each connection has a weight w_{ij}, and the weights are symmetric, so $w_{ij}=w_{ji}$. A unit has no connection to itself. Each unit also has a **bias** b_i, which sets how likely it is to be on.

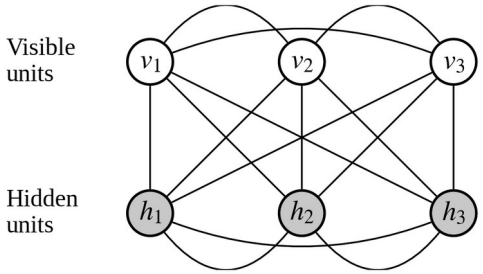


Figure 1: A general Boltzmann Machine with three visible and three hidden units.

2.2 Energy Function

The energy of a state s of the network is

$$E(s) = -\sum_{i<j} w_{ij}\, s_i s_j - \sum_i b_i\, s_i \tag{1}$$

where s_i is the state of unit i, w_{ij} is the weight between units i and j, and b_i is the bias of unit i. The sum over $i<j$ counts each pair once. The state probability is

$$P(s) = \frac{e^{-E(s)}}{Z}, \qquad Z = \sum_{s'} e^{-E(s')} \tag{2}$$

where Z is the partition function, the sum of e^{-E} over all states s', which makes the probabilities add up to one. Lower energy gives higher probability.

2.3 Probabilistic Activation

Units are updated one at a time. Unit i first finds its total input $\Delta E_i=b_i+\Sigma_j\, w_{ij}s_j$, which is the drop in energy when the unit is switched on. It is then turned on with probability

$$P(s_i = 1) = \sigma(\Delta E_i) = \frac{1}{1+e^{-\Delta E_i}} \tag{3}$$

where σ is the logistic sigmoid function: a large positive input almost always turns the unit on, a large negative input almost always turns it off, and zero input gives a probability of 0.5.

2.4 Training Process

Training makes the model distribution match the training data. The weights change as

$$\Delta w_{ij} = \eta\left(\langle s_i s_j\rangle_{\text{data}} - \langle s_i s_j\rangle_{\text{model}}\right) \tag{4}$$

where η is the learning rate, $\langle s_i\, s_j\rangle_{data}$ is the average product of two unit states when the visible units are fixed to training examples, and $\langle s_i\, s_j\rangle_{model}$ is the same average when the network runs freely. Biases use $\langle s_i\rangle$ in the same way. Data correlations are strengthened and the network's own correlations are weakened. The model term needs long sampling, so training is slow.

3. Types, Applications, Advantages and Limitations

3.1 Types

General Boltzmann Machine. Every unit is connected to every other unit, including connections within the visible group and within the hidden group. It is the most flexible form, but learning is very slow.

Restricted Boltzmann Machine (RBM). An RBM has one visible layer and one hidden layer. Units in different layers are connected, but there are no connections within a layer. Its energy is

$$E(v,h) = -\sum_i a_i v_i - \sum_j c_j h_j - \sum_{i,j} v_i w_{ij} h_j \tag{5}$$

where a_i and c_j are the biases of visible unit i and hidden unit j, and w_{ij} is the weight between them. Because of the restriction, all hidden units can be sampled in parallel given the visible units, and the reverse is also true. RBMs are commonly trained with Contrastive Divergence (Hinton, 2002), an approximate method, though it is not the only one.

Deep Boltzmann Machine (DBM). A DBM has several hidden layers stacked above the visible layer, with undirected connections only between neighbouring layers. Higher layers can learn more abstract features (Salakhutdinov and Hinton, 2009).

3.2 Applications

Feature learning: the hidden units learn features of the input without labels, and these features can be given to a classifier or used to initialise a deep network. **Recommendation systems:** an RBM can model user–item ratings and predict the missing ones (Salakhutdinov et al., 2007). **Pattern recognition:** features learned from images, such as handwritten digits, can help a classifier to recognise patterns (Salakhutdinov and Hinton, 2009). **Generative modelling:** after training, the sampling process can produce new samples that resemble the training data. **Dimensionality reduction:** a hidden layer with fewer units than the visible layer gives a compact code of the input. These uses are mostly of academic and historical importance, and many newer methods now perform better.

3.3 Advantages

Boltzmann Machines can learn a probability distribution over the data, so they can also be used to generate samples. They can discover hidden patterns in data without labels, which makes them useful for unsupervised representation learning.

3.4 Limitations

Training is computationally expensive because it needs a lot of sampling. The partition function Z sums over all possible states, so it is hard to compute and model probabilities are difficult to estimate. Training is complex and may converge slowly or be unstable. Compared with many modern deep learning methods, their practical use is limited.

3.5 Example: Learning User Preferences with an RBM

Let each visible unit stand for a movie, with 1 if a user liked it and 0 otherwise, so each user is one training example. The hidden units are not labelled, but after training with Contrastive Divergence, one hidden unit may switch on for users who liked several science-fiction films, because those films are often liked together. For a new user, the hidden units are computed from the movies already liked, and they are then used to find the probability of each unseen movie. A high probability suggests a possible recommendation. This is only an illustration; no experiment was carried out.

4. Conclusion

A Boltzmann Machine is a stochastic neural network of binary visible and hidden units joined by symmetric weights and influenced by bias values. Its behaviour is described by an energy function, and the probability of each state is obtained from this energy using the Boltzmann distribution. Units are switched on or off with a probability that depends on their total input. Learning changes the weights by comparing the correlations measured when the network is driven by data with those it produces by itself, so that the training data receive low energy. The general model is very slow to train, so the Restricted Boltzmann Machine, usually trained with Contrastive Divergence, and the Deep Boltzmann Machine were developed. Boltzmann Machines are significant because they showed how a network can learn a probability distribution and useful features from unlabelled data. They have been applied to feature learning, recommendation, pattern recognition, generative modelling and dimensionality reduction. However, training is expensive, probabilities are hard to estimate, and many modern deep learning methods are easier to train and often perform better in practice. Even so, studying Boltzmann Machines gives a good foundation for energy-based models and generative learning.

Table 1: Summary of the three types of Boltzmann Machine.

Type	Connections	Layers	Training
General Boltzmann Machine	Between all pairs of units, including units of the same group	Visible and hidden units, with no layer structure	Very slow; needs long sampling
Restricted Boltzmann Machine	Only between visible and hidden units	One visible layer and one hidden layer	Approximate methods such as Contrastive Divergence
Deep Boltzmann Machine	Only between neighbouring layers	One visible layer and several hidden layers	Approximate learning procedure of Salakhutdinov and Hinton (2009)

This report is based on a study of published literature. No model was implemented and no experimental results are reported.

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